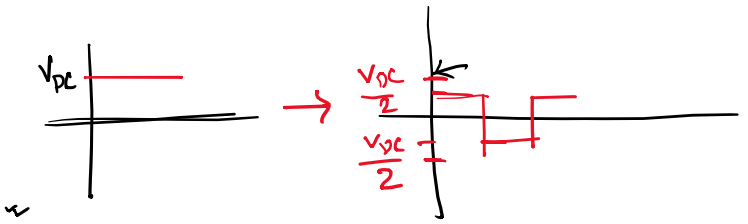
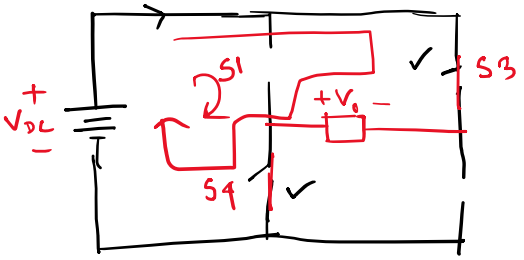
Single phase Full Bridge Inverter

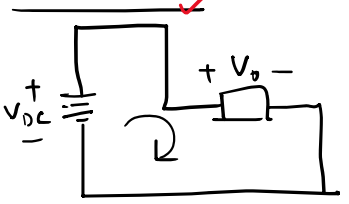
Wednesday, April 29, 2026 3:40 PM



##



S1, S6 → ON S3, S4 → OFF



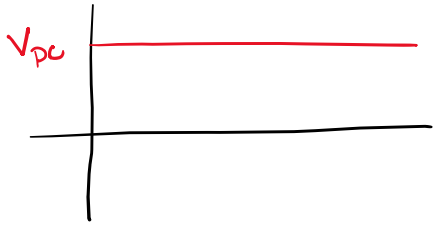
$-V_{DC} + V_o = 0$

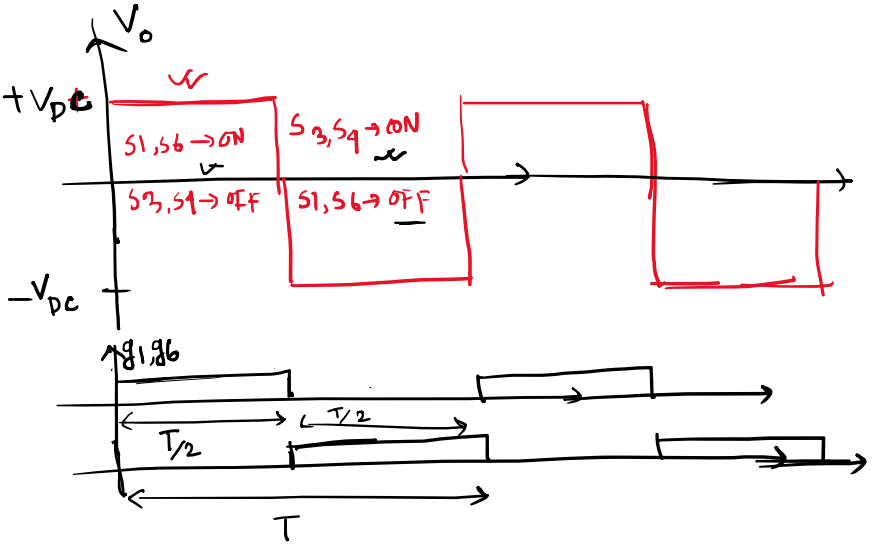
$\Rightarrow V_o = V_{DC}$

S1, S6 → OFF S3, S4 → ON

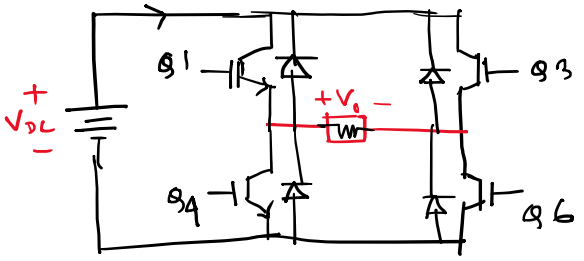
$-V_{DC} - V_o = 0$

$\Rightarrow V_o = -V_{DC}$





##



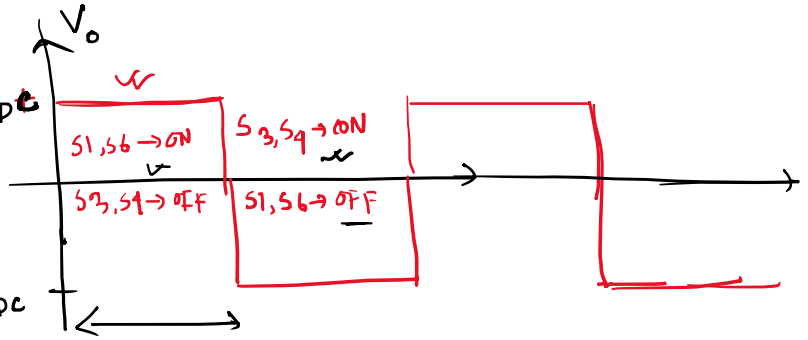
##

V_o

$$V_{o,rms}^2 = \frac{1}{2\pi}\left[\int_0^{\pi} V_{DC}^2\, d\theta + \int_{\pi}^{2\pi} V_{DC}^2\, d\theta\right]$$

$$= \frac{1}{2\pi}\left(V_{DC}^2 \cdot \pi + V_{DC}^2 \cdot \pi\right)$$

$$= \frac{1}{2\pi} \cdot V_{DC}^2 \cdot 2\pi$$



$$\boxed{V_{o,rms} = V_{DC}}$$

odd signal

$$V_o = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t + b_n \sin n\omega t \quad \longrightarrow$$

$a_0, a_n \rightarrow 0$

$$b_n = \frac{1}{T/4}\int_0^{T/2} x(t) \sin n\omega t \; d(\omega t)$$

$$= \frac{4}{2\pi}\int_0^{\pi} V_{DC} \sin(n\theta)\, d\theta$$

$$= \frac{2V_{DC}}{\pi} \cdot \int_0^{\pi} \sin n\theta \, d\theta$$

$$= \frac{2V_{DC}}{n\pi}\left[-\cos n\theta\right]_0^{\pi}$$

$$= \frac{2V_{DC}}{n\pi}\left[1 - (\cos n\pi)\right]$$

$$= \begin{cases} 0 & n = \text{even} \\ \frac{4V_{DC}}{n\pi} & n = \text{odd} \end{cases}$$

$$V_o = \sum \frac{4V_{DC}}{n\pi} \sin(n\omega t) = \sum_{n=1,3,5}^{\infty} \frac{4V_{DC}}{n\pi} \sin(n\omega t)$$

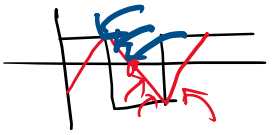
$$V_o = \sum_{n=odd} \frac{4V_{DC}}{n\pi} \sin(n\omega t) = \sum_{n=1,3,5} \frac{4V_{DC}}{n\pi} \sin(n\omega t)$$

$n = 1$

$$V_{o1} = \boxed{\frac{4V_{DC}}{\pi} \sin(\omega t)} \leftarrow$$

$$V_{o1,rms} = \frac{4V_{DC}}{\sqrt{2}\,\pi} = 0.90\,V_{DC}$$

HF, DF, THD = ?

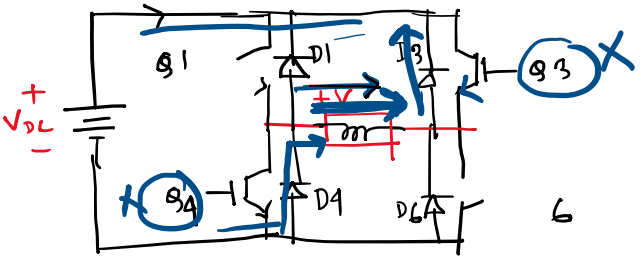


Load → L

S1, S6 → ON S3, S4 → OFF

S3, S4 → ~~ON~~ S1, S6 → OFF

D3, D4 → Feedback Diodes



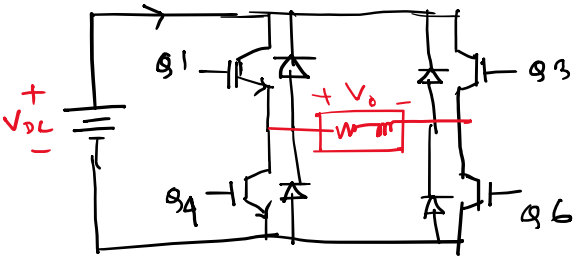
S3, S4 → ON S1, S6 → OFF

S1, S6 → ON

D1, D6 → Feedback diodes

Load → RL

$$V_o = \sum_{n=1,3,5}^{\infty} \frac{4V_{DC}}{n\pi} \sin(n\omega t)$$



$$i_o = \frac{V_o}{Z} = \sum_{n=1,3,5}^{\infty} \frac{4V_{DC}}{n\pi\sqrt{R^2+(n\omega L)^2}} \sin(n\omega t - \theta_n)$$

$$Z = R + j(n\omega L)$$

$$Z = \sqrt{R^2+(n\omega L)^2} \angle \tan^{-1}\left(\frac{n\omega L}{R}\right)$$

θ_n

$n = 1$

$$|i_{o1}| = \frac{4V_{DC}}{\pi\sqrt{R^2+(\omega L)^2}}$$

$$i_{o1,rms} = \frac{4V_{DC}}{\sqrt{2}\pi\sqrt{R^2+(\omega L)^2}}$$

$$P = i_{o1,rms}^2 \cdot R$$